

Listening Without Labels: Open-Set Acoustic Threat Detection for Forest Conservation from Background Audio Alone

Anonymous submission

Abstract

Acoustic monitoring is a scalable defense against illegal logging, but standard detectors, closed-set classifiers trained on public audio, fail on new forest sites. Using an audited, leakage-free, leave-one-site-out protocol across four rainforest locations, we show the breakdown is not in discrimination but in operating-point transfer: a classifier at 0.907 in-domain accuracy detects essentially no chainsaws on any held-out site, even where its own score still ranks chainsaws above background at ROC-AUC 0.964. A linear probe recovers recording site from the learned representation at $4\times$ chance and more strongly than it recovers event class, giving a mechanism for the failure. We propose a background-first, open-set detector that sidesteps the trap: it fits a density over a frozen general-purpose embedding on unlabeled deployment-site background, calibrates a threshold to an explicit background false-positive budget, and attributes flagged events with per-class prototypes built incrementally from ranger-verified positives, reserving honest *unknown* mass otherwise. On the primary held-out site it reaches AUC 0.713 and flags 0.317 of chainsaws at a 0.056 false-positive rate, exceeding the calibrated closed-set recall of 0.195 while using zero labeled positives to decide what is anomalous. A supervised probe on the same frozen embeddings prices what label-free operation costs and isolates the binding constraint as representational: weighted AUROC is 0.783 for AudioSet and 0.786 for a bird-specific encoder (BirdNET), but only 0.648 for the site-trained representation. Deployed in an open-source monitoring platform we release, ranger feedback closes the loop, with each verified alert becoming a prototype update and no retraining. We release code, fitted models, and the full protocol.

1 Introduction

Illegal logging is a leading driver of tropical forest loss, with organized criminal activity implicated in a large share of it (Nellemann 2012), and its most reliable early signature is sound: chainsaws and vehicles are audible under canopy that hides the activity from optical satellites until after the trees are down. Passive acoustic monitoring has consequently become a standard conservation instrument, to the point that

its bottleneck has shifted from collecting audio to interpreting it (Stowell 2022; Malerba et al. 2026). The dominant recipe is a closed-set classifier: train a CNN on labeled clips of k threat classes, deploy, and alert when a class probability crosses a threshold. This is not a strawman we construct: it is the recipe fielded conservation-acoustic systems actually run, and a global expert assessment of the field identifies manual validation of detector output as a standing bottleneck, since practitioners do not act on raw detections unchecked (Malerba et al. 2026). The failure mode we measure is thus a property of the deployed paradigm, not of a weak baseline.

This recipe embeds two assumptions that fail in conservation deployments. First, it assumes labeled positives from the deployment domain exist. They rarely do: a new reserve has no verified recordings of chainsaws *under its own microphones, weather, and species chorus*, and public datasets are dominated by close-range, high-SNR urban recordings. Label scarcity is in fact the standing premise of the bioacoustics literature, which is why few-shot formulations are the field’s dominant response (Stowell 2022; Nolasco et al. 2023). Second, it assumes the threat taxonomy is known in advance, so any sound outside the k classes is silently mapped onto them or suppressed.

We first quantify how badly the closed-set recipe travels, and diagnose *why* (Section 3). Across four held-out rainforest sites, with leakage-checked splits and validation-calibrated thresholds, we find three things. **(1) Random splits are optimistic:** a classifier at 0.907 in-domain accuracy detects essentially no chainsaws on any held-out site. This is the acoustic instance of a known hazard in spatially structured ecological data, where random cross-validation ignores dependence between nearby samples and underestimates predictive error, and block or grouped splits are the corrective (Roberts et al. 2017). **(2) The failure is operating-point transfer, not discrimination:** the score still ranks chainsaws above background (ROC-AUC up to 0.96), but the source-calibrated threshold yields 0 recall in the target domain, so false-positive control at a fixed budget, not accuracy, is the binding constraint. That a model’s scores decalibrate under distribution shift even while its ranking survives is a documented failure mode (Snoek et al. 2019); our contribution is to show that it, rather than discrimination, is what actually blocks deployment here. **(3) The representation encodes site:** recording site is linearly decodable from the embedding

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far above chance, and more strongly than event class, giving a mechanism for the failure. Sensor- and site-dependent background structure is known to break bioacoustic detectors moved to a new recording unit (Lostanlen et al. 2019); we show it is legible directly in the representation, before it shows up in any metric. A CNN trained on rainforest soundscapes with a site held out confirms the pattern, reaching only 0.195 recall at a deployable false-positive rate despite 0.976 raw-argmax recall. Our open-set detector, over a frozen general-purpose embedding, recovers a deployable operating point on the same held-out site, reaching 0.317 flagged recall at 0.056 background FP, exceeding the closed-set recall while using no labeled positives to decide what is anomalous.

We argue the deployable formulation is open-set and background-first:

1. **Flag** a clip by how far it deviates from *this forest's* normal soundscape, a density fit on unlabeled background audio alone, which every deployment can record on day one;
2. **Attribute** flagged clips against per-class prototypes built from whatever ranger-verified positives exist so far, zero at launch;
3. **Abstain honestly**: when no prototype matches, report *unknown* rather than forcing a known label.

The decision threshold is calibrated to an explicit background false-positive budget, because ranger attention, not accuracy, is the scarce resource.

Our contributions:

- A diagnosis of closed-set failure under site-level shift with an audited, leakage-free, multi-site protocol (Section 3), separating three causes with bootstrap-quantified evidence: split optimism, an operating-point (not discrimination) failure in which false-positive control is the binding constraint, and a representation that encodes recording site more strongly than event class. To our knowledge this is the first study to attribute bioacoustic threat-detection failure to site-encoding via a linear-probe analysis.
- A background-only open-set detector, built from a shrunk-covariance Mahalanobis density over a frozen pretrained embedding, quantile-mapped scores, an FP-budget threshold, and prototype attribution with reserved unknown mass, that ships with zero labeled threats and improves monotonically as verified positives arrive, without retraining.
- A site-held-out evaluation across four RFCx sites measuring flagging, attribution, the attribution-vs-verified-positives curve, and honest-unknown behavior under class holdout. The embedding, not the detector head, governs cross-site deployability: frozen general-purpose encoders clear chance separability at every site where the site-trained representation does not.
- A measurement of what label-free operation costs. A supervised probe on the *same* frozen embeddings, under the identical protocol, gives the label-rich upper bound per representation (weighted AUROC 0.783 AudioSet, 0.786 BirdNET, 0.648 site-trained). Labels roughly double recall, but the label-free detector is the one that runs on day one. We also show an embedding can rank well and still

be undeployable: BirdNET is the best probe substrate yet flags nothing under a transferred threshold, which is the same operating-point failure in unsupervised form.

- An open-source deployment in which the ranger review loop closes the label flywheel: confirmed alerts become prototype updates.

2 Related Work

Acoustic monitoring for conservation. Passive acoustic monitoring is established for biodiversity and increasingly for threat detection (chainsaws, gunshots, vehicles), with deployments such as Rainforest Connection converting recordings into ranger alerts (Stowell 2022). These systems predominantly use closed-set supervised classifiers and report in-domain metrics; cross-site generalization under a fixed false-positive budget is rarely measured. A global expert assessment of PAM practice identifies manual validation of detector output as a priority bottleneck (Malerba et al. 2026), which is why we treat reviewer attention, not accuracy, as the operative constraint. Our negative result quantifies the cross-site gap this reflects, and our detector targets the label-scarce, open-taxonomy regime deployments actually face.

Few-shot bioacoustic event detection. The DCASE few-shot bioacoustic task formalizes detecting novel sound events from a handful of labeled examples (Nolasco et al. 2023). Our prototype stage is in this spirit, since a class is defined by a few verified positives, but we couple it to a background-only anomaly gate and an explicit unknown option, so the system flags before any labels exist and abstains when no prototype matches.

Open-set recognition and OOD detection. Open-set recognition asks a classifier to reject inputs outside its training classes; a large OOD-detection literature scores novelty, including Mahalanobis distance in a deep feature space (Lee et al. 2018). We adopt the Mahalanobis-in-embedding scoring but invert the usual framing: the “in-distribution” density is *normal forest background*, not the labeled classes, so the detector is trained without any positives and known classes enter only as post-hoc prototypes.

Prototypical and metric learning. Class-mean prototypes with cosine similarity are standard in few-shot metric learning (Snell, Swersky, and Zemel 2017). Our contribution is not the prototype but its role in a deployable pipeline: prototypes are optional, incrementally updatable from human confirmations, and gated behind a background-calibrated anomaly threshold with reserved unknown mass.

Transfer of pretrained audio embeddings. Large audio taggers pretrained on AudioSet (Gemmeke et al. 2017), such as CNN14 (Kong et al. 2020), yield general-purpose embeddings that transfer across audio tasks, as does the bioacoustics-specific BirdNET (Kahl et al. 2021). We use one as a frozen ϕ and show that the choice of embedding, not the detector head, governs cross-site deployability for forest threats.

Distribution shift, shortcut learning, and evaluation optimism. Models often exploit spurious, environment-correlated features rather than the signal of interest (Geirhos et al. 2020), and random train/test splits that share those confounds overstate generalization; sensor- and site-dependent background structure is a documented confound in bioacoustics (Lostanlen et al. 2019), and random cross-validation over spatially structured ecological data is known to underestimate error (Roberts et al. 2017). Our diagnosis instantiates this for forest threat detection: a leakage-free but random split hides a cross-site collapse, and a linear probe shows the representation encodes site identity. We use leave-one-site-out evaluation and false-positive-budgeted recall to measure what random splits and accuracy conceal.

3 Diagnosis: Why Closed-Set Classifiers Fail Under Site Shift

Before proposing an alternative, we establish *why* the closed-set recipe fails, testing three hypotheses:

- H1 (split optimism)** in-domain (random-split) metrics substantially overstate cross-site deployment performance;
- H2 (FP-control is binding)** cross-site failure is an operating-point failure, not a discrimination failure: the score often still ranks threats above background, but no source-calibrated threshold is usable;
- H3 (site encoding)** the learned representation encodes recording *site* as strongly as event *class*, giving a mechanism for the failure to transfer.

Setup (uncontaminated by construction). Our public-trained models contain *zero* clips from the RFCx rainforest sites (verified over the training manifests). Evaluating them on those sites is therefore a clean public→forest transfer test requiring no retraining, pure inference. We use the four RFCx sites with enough chainsaw and background clips for a per-site estimate: warsi (412/198), tambopata (41/72), romania (30/15), pooks (10/12). Each model is applied with its own validation-calibrated deployment thresholds (deploy-what-you-calibrated). All metrics carry clip-level bootstrap 95% CIs (1000 resamples). Models that trained on RFCx sites are excluded from cross-site claims to avoid contamination.

H1: in-domain metrics are optimistic. Random splits over spatially structured data are known to underestimate predictive error (Roberts et al. 2017); we quantify the size of that gap here. On its own held-out (but same-domain) test split, the production classifier scores 0.907 accuracy, but this is imbalance inflation (the split is vehicle-dominated); macro-F1 is 0.540 and the raw-argmax background threat-false-positive rate is 0.556 (more than half of background alarms). Carried to any RFCx site, calibrated chainsaw recall is ≈ 0 (Table 1). A random split thus overstates deployment by the entire usable range of the metric.

H2: the collapse is universal and is a calibration failure. Table 1 shows deployment-calibrated chainsaw recall is statistically indistinguishable from zero at *every* site: the failure is not specific to one unusual site. Yet the threshold-free chainsaw-vs-background ROC-AUC is well above chance at

Table 1: Public-trained classifier (never trained on any RFCx clip) on four held-out RFCx sites, deployment-calibrated, with bootstrap 95% CIs. Recall is ≈ 0 everywhere, but chainsaw-vs-background ROC-AUC is well above chance (decisively so on romania (0.964) where recall is still 0.000). The failure is one of operating-point transfer, not discrimination.

Held-out site	Chainsaw recall	Background FP	Score AUC
tambopata	0.000 [.00,.00]	0.014 [.00,.05]	0.559 [.45,.67]
warsi	0.002 [.00,.01]	0.066 [.03,.10]	0.652 [.61,.70]
romania	0.000 [.00,.00]	0.000 [.00,.00]	0.964 [.91,1.0]
pooks	0.000 [.00,.00]	0.167 [.00,.39]	0.725 [.47,.92]

the same sites, reaching 0.964 on romania *where recall is exactly* 0.000. The classifier can rank chainsaws above background; the source-calibrated decision threshold simply does not transfer to the target domain, matching the broader finding that scores decalibrate under distribution shift (Snoek et al. 2019). Accuracy and AUC hide this; recall-at-a-fixed-FP-budget exposes it. False-positive control, not raw discrimination, is the binding constraint. This is why we budget false positives rather than optimize accuracy: in deployment the scarce resource is ranger attention, and operators already triage only a subset of alerts because raw detections are untrusted. The operative quantity is therefore recall at a *reviewable alert volume*, the number of alerts a patrol team can act on per day, for which a background false-positive budget is a direct proxy. A detector that floods that budget (up to 0.85 background-FP here) is unusable regardless of its ranking quality.

H2, cont.: training interventions do not fix it. Table 2 walks the deployed training lineage as an intervention comparison on the highest-support site (warsi), each recipe under its own calibrated thresholds. Hard-negative mining is the *only* intervention that meaningfully recovers cross-site recall (0.495) and gives the best separability (AUC 0.720), but at a 0.136 background-FP, above the 0.10 budget. Every other recipe either detects almost nothing or floods (up to 0.854 FP). No public-data recipe is simultaneously sensitive and FP-compliant across the domain gap. (The recipes are not strictly nested supersets; we read this as a lineage comparison, not a perfectly controlled ablation.)

H3: the representation encodes site, not just event. To probe the mechanism, we fit two linear classifiers (logistic regression, 5-fold stratified CV, balanced accuracy) on the *same* frozen embeddings of 429 forest clips from 7 sites, clips these models never saw in training. If site is as decodable as class, the representation is organized by *where* a clip was recorded, not *what* it contains. Table 3 confirms it: recording site is linearly decodable from the supervised-CNN embedding at 0.600 balanced accuracy ($\approx 4 \times$ its 0.143 chance), and more strongly than from a frozen general-purpose encoder (0.508), while class-discrimination is essentially tied. A network that never trained on forest audio still geometrically organizes forest clips by site, consistent with reports that sensor- and site-specific background structure degrades bioacoustic de-

Table 2: Training-recipe comparison on held-out warsi, deployment-calibrated. Hard negatives are the only lever that recovers cross-site recall, and even they exceed the 0.10 FP budget.

Training recipe	Recall	Bg FP	AUC
baseline (balanced public)	0.005	0.030	0.580
+ augmentation	0.002	0.066	0.652
+ hard negatives	0.495	0.136	0.720
+ more hard negatives	0.012	0.197	0.682
+ expanded set	0.000	0.854	0.654
+ DCASE machine negatives	0.022	0.556	0.632
+ conservative background	0.000	0.040	0.661

Table 3: Linear-probe balanced accuracy on identical frozen embeddings of held-out forest clips (429 clips, 7 sites). Recording site is decoded far above chance, more so by the supervised CNN, while class accuracy is tied. The supervised representation is disproportionately organized by site.

Embedding	Predict class (chance 0.50)	Predict site (chance 0.14)
Supervised forest CNN	0.854	0.600
Frozen AudioSet CNN14	0.839	0.508

tectors moved to a new recording unit (Lostanlen et al. 2019) (Figure 1): spurious site structure the closed-set head cannot help but exploit, and which does not transfer to a new site.

Takeaway. Closed-set classifiers fail under site-level shift not because they cannot discriminate threats, but because (i) their operating point, calibrated on the source domain, does not transfer, and (ii) their representation encodes site-specific structure. This motivates a detector whose decision boundary is defined by the *deployment site’s own* background and whose attribution is decoupled from a frozen, general-purpose representation, the design of Section 4.

4 Method

Let $\phi: \mathcal{X} \rightarrow \mathbb{R}^d$ be a fixed embedding of an audio clip. We deliberately *freeze* ϕ : the detector never backpropagates into it, so a new class or a new site never triggers retraining. All learning happens in two lightweight, closed-form stages over embeddings.

4.1 Stage 1: Background density and flagging

Given embeddings $\{\phi(x_i)\}$ of unlabeled forest-at-rest clips, we fit a Gaussian with a shrunk covariance for a well-conditioned inverse (Ledoit and Wolf 2004):

$$\hat{\Sigma} = (1 - \alpha) S + \alpha \text{diag}(S) + \epsilon I, \quad (1)$$

where S is the sample covariance and α the shrinkage. A clip’s raw anomaly is its Mahalanobis distance (Lee et al. 2018) $r(x) = \sqrt{(\phi(x) - \hat{\mu})^\top \hat{\Sigma}^{-1} (\phi(x) - \hat{\mu})}$. Raw distances are not comparable across sites, so we map them through the *empirical background CDF*: the anomaly score

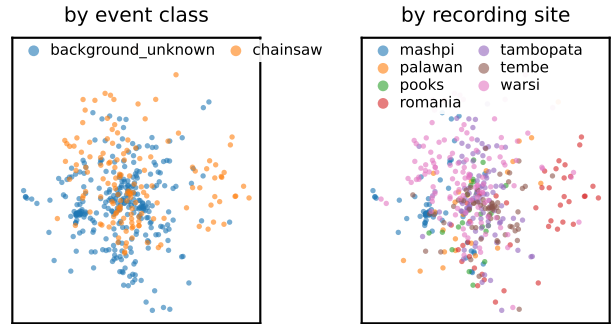


Figure 1: PCA of the supervised-CNN embedding of held-out forest clips, colored by event class (left) and by recording site (right). Structure tracks site at least as strongly as class.

$s(x) \in [0, 1]$ is the quantile of $r(x)$ among training-background distances, so $s = 0.99$ means “more unusual than 99% of this forest’s background.”

FP-budget calibration. We flag when $s(x) \geq \tau$. Rather than a fixed τ , we pick the smallest threshold whose background false-positive rate on a held-out background split is at most a budget β (we use $\beta = 0.10$, the platform’s promotion gate). This maximizes recall subject to the ranger-attention budget and is the only quantity a site operator must set. Crucially, Stage 1 needs *no labeled positives*.

4.2 Stage 2: Open-set attribution

For each known class c with verified positives, we store an L_2 -normalized mean prototype $p_c = \widehat{\text{mean}}_i \phi(x_i)$. For a flagged clip we compute cosine similarities $\sigma_c = p_c^\top \hat{\phi}(x)$ and split the probability mass into a known part and an honest unknown part. Letting $\sigma^* = \max_c \sigma_c$ and a similarity floor θ , the known mass is

$$m = \text{clip}\left(\frac{\sigma^* - \theta}{1 - \theta}, 0, 1\right), \quad (2)$$

distributed over classes by a temperature-softmax of $\{\sigma_c\}$, with the remaining $1 - m$ assigned to *unknown*. With zero prototypes, $m = 0$ and every anomaly is honestly unknown, and flagging is unaffected. As verified positives arrive for a class, its prototype sharpens and its mass grows, *monotonically and without retraining*, which is exactly the incremental behavior a ranger-in-the-loop deployment needs.

4.3 Embedding choice

ϕ is pluggable. We compare a site-trained CNN’s pre-classifier feature (64-d) against two frozen general-purpose encoders: AudioSet-pretrained CNN14 (2048-d) and BirdNET V2.4 (1024-d), a bioacoustics encoder trained for bird vocalization. All three are evaluated under identical background-density and prototype machinery. Because Stage 1 and Stage 2 are closed-form over embeddings, swapping ϕ is the only change required to test a new representation, which is what lets us attribute cross-site performance to the representation rather than the detector.

5 Experiments

Data and protocol. We use RFCx rainforest soundscapes (Rainforest Connection 2024) with chainsaw and background labels from four sites carrying enough of both for a per-site estimate: warsi (412 chainsaw / 198 background test clips), tambopata (41/72), romania (30/15), pooks (10/12). Every experiment is *leave-one-site-out*: all clips from the target site are reserved for test, everything is fit on the remaining sites, and the decision threshold is calibrated on a training-site background validation split at a background false-positive budget of 0.10. A recording-level audit confirms zero clips from the held-out site in train or validation and zero shared source recordings across splits, so no result below is contaminated by clip-window leakage.

Metrics. *Flagged recall* (positives scoring above threshold), *attributed recall* (flagged *and* the predicted kind is correct), *background FP*, and threshold-free AUC of the score separating chainsaw from background. For the supervised probe we additionally report *event-level recall*: clips are grouped back into source recordings and an event counts as detected if any of its clips fires, since a chainsaw runs continuously and the operational unit is the event, not the 3-second clip.

Implementation and reproducibility. The detector and evaluation math are NumPy and unit-tested (123 tests); audio inference uses PyTorch 2.7 with `panns_inference` (AudioSet CNN14, 2048-d) and `birdnetlib` (BirdNET V2.4, 1024-d) on CPU/Apple-Silicon under macOS. Randomized results fix seeds and every interval is a $1000\times$ bootstrap. All numbers below were regenerated from scratch for this submission; code, fitted artifacts, manifests, and the exact commands are released. The evaluation audio is public and released under CC BY-NC 4.0; the closed-set lineage additionally trains on public forest and urban corpora (Bandara et al. 2023; Gemmeke et al. 2017).

5.1 From diagnosis to a deployable operating point

Section 3 established that closed-set classifiers fail across sites through operating-point transfer and site encoding. Table 4 compares, on the primary held-out site, the strongest closed-set baseline against the open-set detector under two embeddings. The closed-set model is a CNN trained on rainforest audio *with tambopata held out*, the most favorable closed-set setting: 0.976 chainsaw recall at raw argmax, but with half of all background flagged as a threat; calibrated to a deployable false-positive rate, recall collapses to 0.195. Re-using that same CNN as the open-set *embedding* inherits the site bias, giving an anomaly-score AUC of 0.319, below chance, and consistent with the representation analysis in Section 3. Swapping in a frozen AudioSet encoder is what recovers a usable operating point.

5.2 The representation, not the head, governs transfer

Because the open-set machinery is identical across embeddings, the comparison isolates the effect of the frozen encoder

Table 4: Held-out-site (tambopata) chainsaw detection, threshold calibrated to background $FP \leq 0.10$ on training-site validation. All rows are our own re-run under one protocol. “Score AUC” is threshold-free separability *before* any labels are used.

Method	Recall	Bg FP	Score AUC
Closed-set CNN (raw argmax)	0.976	0.500	—
Closed-set CNN (calibrated)	0.195	0.000	—
Open-set, site-trained CNN	0.049	0.028	0.319
Open-set, AudioSet CNN14	0.317	0.056	0.713

Table 5: Leave-one-site-out folds, open-set (background-only) detector. The AudioSet embedding lifts separability above chance at every site where the site-trained CNN is at or below it, but the validation-calibrated threshold does not always hold the FP budget on an unseen site. pooks has only 10 held-out chainsaw clips and should be read as indicative.

Site	n_{ch}	site-trained CNN		AudioSet CNN14	
		Recall	AUC	Recall	AUC
tambopata	41	0.049	0.319	0.317	0.713
warsi	412	0.250	0.481	0.367	0.619
romania	30	0.167	0.564	0.267	0.559
pooks	10	0.000	0.146	0.000	0.546

ϕ . The site-trained CNN inherits the domain bias that sinks the closed-set model (AUC 0.15–0.56, at or below chance), whereas the frozen AudioSet encoder is above chance at every held-out site (0.55–0.71; Table 5). On the primary site it reaches 0.317 flagged recall at 0.056 background FP, exceeding the closed-set calibrated recall of 0.195 while using *zero* labeled positives to decide what is anomalous.

Two caveats temper this. Absolute recall is modest (0.27–0.37 where the FP budget is met): faint, distant chainsaws under an unseen chorus remain hard. And the threshold calibrated on training-site background does not always hold the budget on an unseen site (warsi 0.20, romania 0.13), a cross-site *calibration* gap distinct from the separability the embedding provides.

5.3 What label-free operation costs

To separate what is lost by using no labels from what is lost to site shift, we fit a supervised probe (logistic regression) on the *same* frozen embeddings using the training-site chainsaw labels a real deployment eventually accumulates, under the identical leave-one-site-out protocol and FP budget (Table 6, Figure 2). This is the label-rich upper bound for a given representation.

Three findings follow. First, the general-purpose encoders are far better substrates than the site-trained CNN: weighted AUROC 0.783 (AudioSet) and 0.786 (BirdNET) against 0.648, and on romania BirdNET reaches AUROC 0.973 with 0.966 event recall at 0.067 FP, an operationally useful detector. Second, the two general encoders are near-tied on AUROC despite very different pretraining (AudioSet tagging versus bird vocalization), which points to generic acoustic

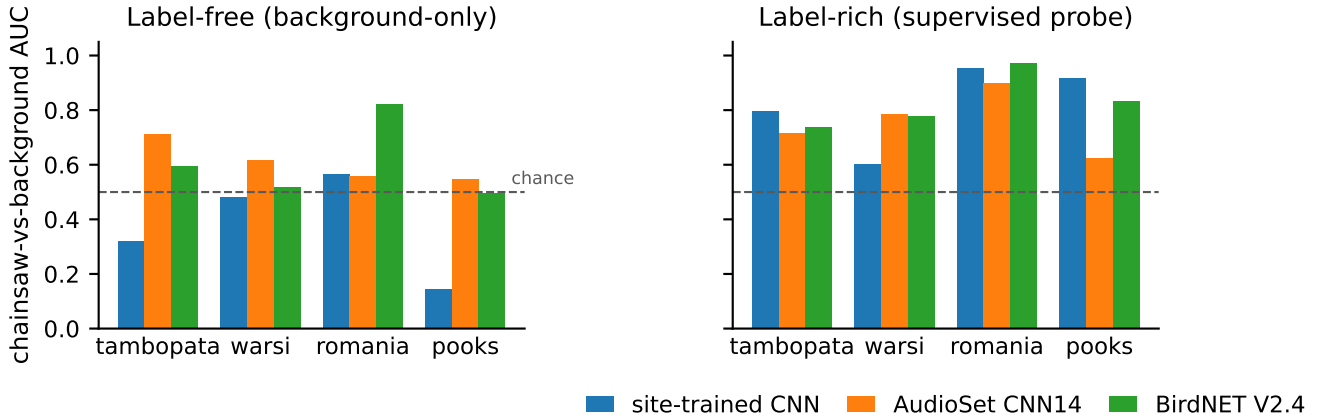


Figure 2: Per-site chainsaw-vs-background AUC, label-free (left) and label-rich (right), for the three frozen embeddings. Dashed line is chance. General-purpose audio representations transfer; the site-trained one does not.

Table 6: Supervised leave-one-site-out probe over frozen embeddings, at the same 0.10 background-FP budget. AUROC is threshold-free, event recall groups clips into source recordings. Rows are n -weighted means over the four folds; per-site AUROC 95% CIs are bootstrap. The choice of representation moves transfer far more than the choice of head.

Frozen embedding	AUROC	Clip recall	Event recall
site-trained CNN	0.648	0.369	0.370
AudioSet CNN14	0.783	0.594	0.595
BirdNET V2.4	0.786	0.527	0.531

generality rather than any threat-specific prior as the active ingredient. Third, comparing Table 6 with Table 5 prices label-free operation: with labels, the AudioSet representation supports 0.594 event recall; without them, background-only flagging on the same representation delivers 0.317 on the primary site. Labels roughly double recall, but the label-free detector is the one that can run on day one.

5.4 Ranking can be strong while calibration fails

BirdNET makes the paper’s central distinction unusually sharp. As a supervised probe it is the best embedding we tested (weighted AUROC 0.786). Yet under the label-free detector it flags *nothing* at every site (recall 0.000), despite threshold-free AUC up to 0.821 on romania. The score ranks chainsaws above background; the background-density threshold calibrated on training sites simply lands in the wrong place on an unseen one. This is the same operating-point failure we diagnosed in the closed-set classifier (Section 3), reappearing in a fully unsupervised detector, and it is why we report recall at a fixed FP budget rather than AUC alone: an embedding can be excellent and still be undeployable if its operating point does not transfer.

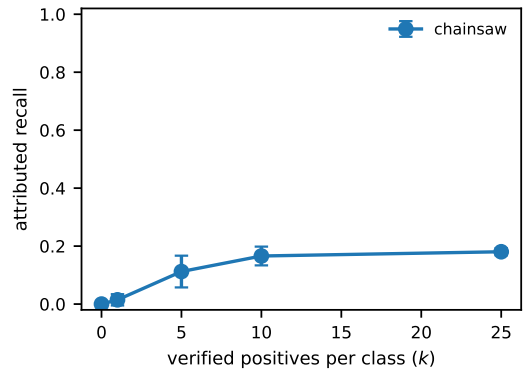


Figure 3: Attributed recall vs. verified positives per class (tambopata held-out, AudioSet embedding; mean $\pm$ std over seeds). Flagging is independent of k ; only attribution depends on it.

5.5 Attribution lights up with a handful of labels

Figure 3 plots attributed recall against the number of verified positives per class $k \in \{0, 1, 5, 10, 25\}$, with flagging held fixed. At $k = 0$ every anomaly is honestly unknown by construction (0.000 attributed recall); on tambopata attribution rises to 0.112 at $k = 5$ and 0.180 at $k = 25$. Flagging is identical across all k by construction, so ranger confirmations only move mass from *unknown* onto the named class, which is the day-one-useful, self-improving behavior a human-in-the-loop deployment needs.

5.6 Honest unknowns

To test abstention we remove chainsaw’s prototype entirely and re-score the held-out site’s chainsaw clips. A well-behaved open-set detector should still *flag* them, since they remain acoustically unusual, while assigning their likelihood mass to *unknown* rather than misattributing it to a remaining class. This is what we observe: flagging is unchanged

and 100% of flagged clips are predicted *unknown* with mean unknown mass 1.0. The detector never invents a label for a class it was never shown.

5.7 Limitations

Our evidence is one threat class (chainsaw) across four rainforest sites, and two folds are small ($n = 30$ and $n = 10$), so per-site numbers carry wide intervals and we weight by n when aggregating. Absolute label-free recall remains modest, and the calibration gap means an operator should expect to spend a small amount of destination-site background to place the threshold. We evaluate on recorded clips, not a live deployment; a field study with in-situ positives is the necessary next step and is underway.

6 Deployment: The Ranger-in-the-Loop Flywheel

The detector ships inside an open-source forest-monitoring platform that we release with this paper (name withheld for anonymous review). A field sensor uploads a clip; the API scores it with the fitted detector and, when $s(x) \geq \tau$, creates an anomaly alert carrying the anomaly score, the predicted kind, and the full likelihood vector including *unknown*. The dashboard renders the ranked breakdown (e.g. “likely chainsaw 71% · vehicle 12% · unknown 17%”) so a ranger sees not just *that* something is unusual but the current best guess and its uncertainty.

Closing the loop without retraining. Ranger confirmations in the clip-review UI are exported as a positives manifest and folded back into the prototype set. Because attribution is closed-form over a frozen embedding, this is a seconds-long refit, not a training run: the very labels the deployment generates are the labels that light up its classes. This is the operational realization of the k -curve in Section 5: each confirmation moves the system rightward along it.

Why background-first fits social-impact constraints. Under-resourced conservation teams cannot commission labeled threat corpora for every reserve, but every site can record its own background on day one. Anchoring the detector to background makes the expensive resource (labeled positives) optional and incremental, and makes the tunable knob a single interpretable quantity, the background false-positive budget that bounds wasted ranger patrols. The honest *unknown* option means a novel threat (a new machine, a new vehicle) is surfaced rather than silently misfiled, which is the failure mode a fixed taxonomy cannot avoid.

Reproducibility. The detector math is a small, dependency-light module unit-tested without the audio stack; the fit and evaluation are single commands over public manifests; and fitted artifacts, the evaluation protocol, and the platform are released open-source. We complete the AAAI reproducibility checklist and document dataset licenses for redistribution.

7 Conclusion

Closed-set acoustic threat classifiers do not survive a change of forest: strong in-domain validation hides a cross-site col-

lapse once a deployable false-positive budget is enforced, and the collapse is one of operating-point transfer rather than discrimination, driven by a representation that encodes recording site more strongly than event class. We reframed the task as background-first and open-set, flagging deviation from a site’s own soundscape, attributing with incrementally built prototypes, and abstaining honestly. The resulting detector ships with zero labeled threats and improves as rangers confirm alerts, without retraining.

Across four held-out sites the choice of frozen representation, not the detector head, governed transfer, and a supervised probe on the same embeddings priced what label-free operation costs: labels roughly double recall, but only the label-free detector can run on day one. BirdNET sharpened the central lesson by being the strongest probe substrate while flagging nothing under a transferred threshold, a reminder that ranking quality and deployability are different properties and that false-positive control is what binds.

Beyond forests, the same recipe fits any monitoring setting where normal is easy to record, anomalies are rare and open-ended, and expert attention is the limiting resource. Limitations remain: one threat class, four sites, two of them small, modest absolute recall, and a calibration gap that a little destination-site background should close. Field validation on in-situ positives is the necessary next step and is underway.

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